



30753 LARGE LANGUAGE MODELS FOR MARKET RESEARCH
Bocconi University

AIRBNB OR LONG-TERM LEASE?

Where does the rental money really go in Milan?

Group 13
I HUI CHEN
Siying Li



You own an apartment in Milan...

Airbnb or Long-term Lease?



- Premium varies from +40% to +141% by location
- No transparent neighbourhood-level comparison exists
- One choice = 10-year income trajectory

OUR SOLUTION

Interactive Dashboard

backed by **Quantitative Analysis** across 87 neighbourhoods

- Neighbourhood-level precision
- Side-by-side comparison
- Interactive map
- Free & accessible



DATA SOURCES

Airbnb Listings

20,553

Cleaned Listings



Original: 22,948 -10.4% dropped

- Inside Airbnb
- ~ March 2026
- Dropped missing price

Long-term Rent

31

Immobiliare.it zones

€15.94–31.34

/m²/month asking rents

Neighbourhoods

87

Total Neighbourhoods

87

31

Manual mapping to rent zones

Full coverage achieved

Time Window

10

Months

June 2025 – March 2026

Consistent window

All zones manually refreshed to
March 2026

METHOD

Step 1: Airbnb Monthly Revenue (Estimated)

$\text{estimated_monthly_revenue} = \text{price} \times (\text{availability_365} / 12)$

Using availability as occupancy proxy; aggregated to median per neighbourhood



Step 2: Long-Term Rent (Standardized 60 m²)

$\text{longterm_monthly_rent} = \text{rent_€_m}^2_\text{month} \times 60$

60 m² assumption for comparison (adjustable in dashboard 40–120 m²)



Step 3: Premium (Percentage)

$\text{premium} = (\text{median_revenue} - \text{longterm_rent}) / \text{longterm_rent}$

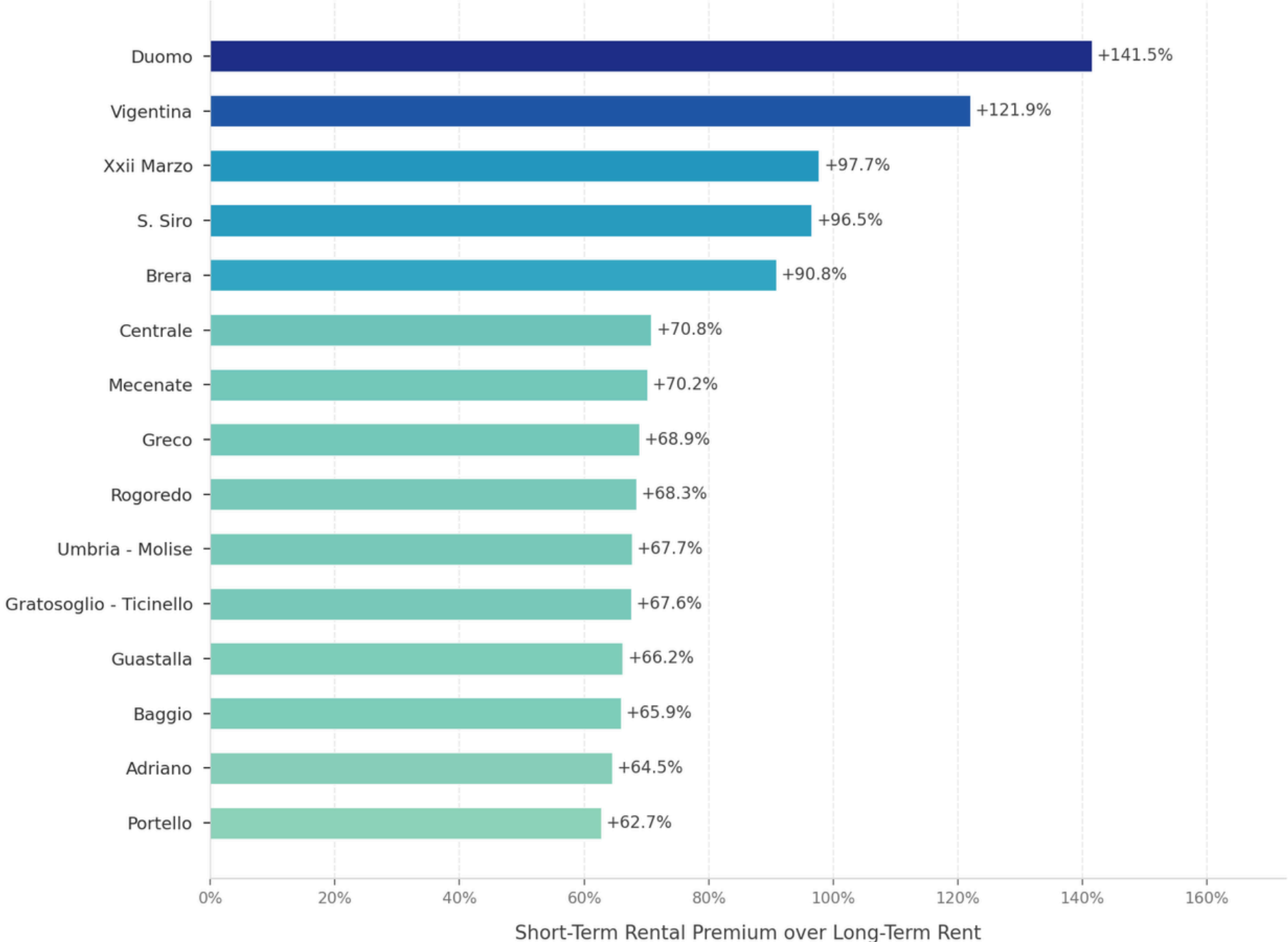
+100% premium = Airbnb earns 2× long-term rent (gross, not net)

 **Note:** Claude Code role: Data cleaning, merge diagnostics, premium calculation, visualization, dashboard development (100% of technical execution)

Chart 1: Top 15 Bar Chart

Top 15 Milan Neighbourhoods by Short-Term Rental Premium

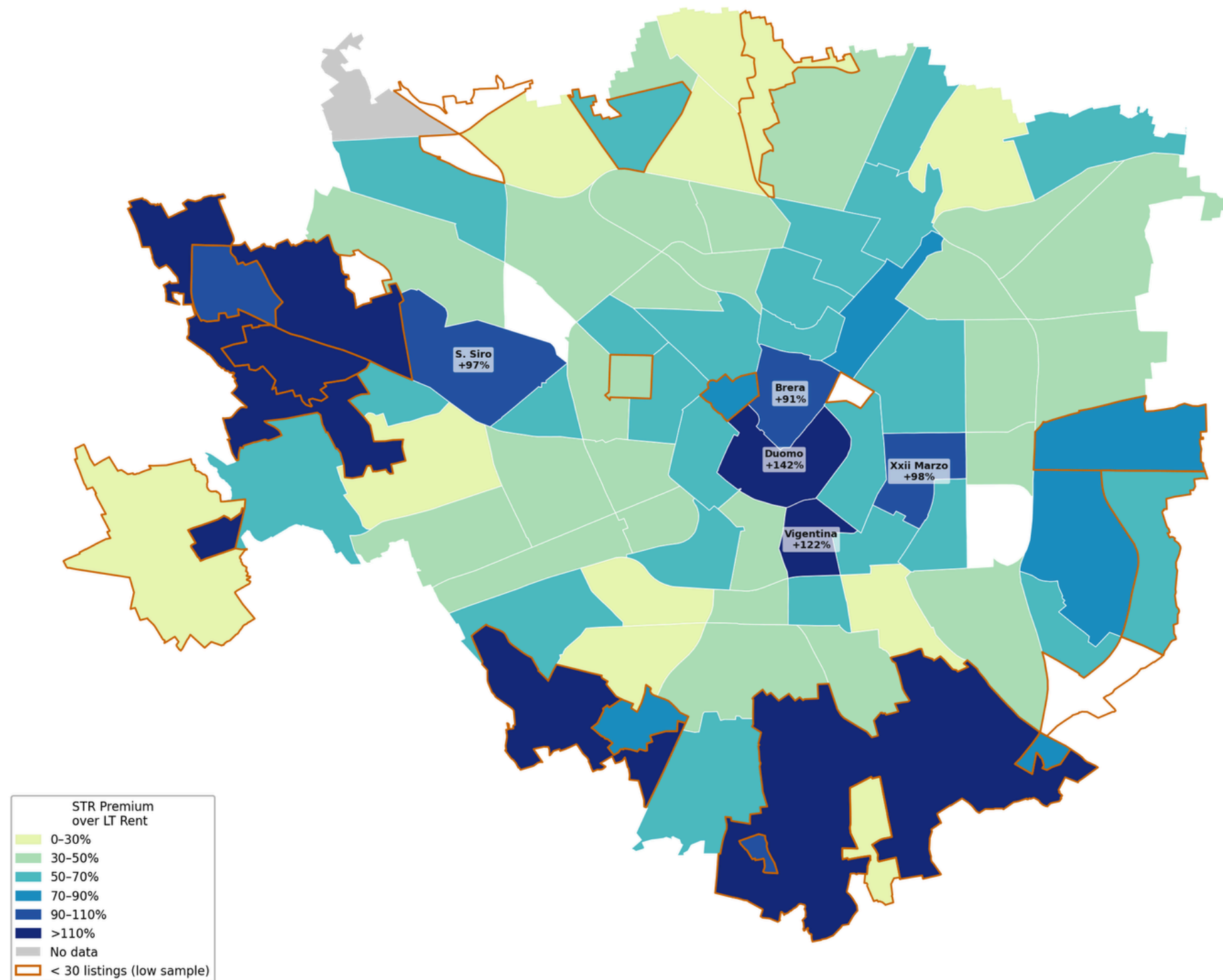
Median Airbnb monthly revenue vs long-term rent · 60 m² apartment · Data window: 2025-06 to 2026-03



- Duomo +141.5% — highest rent AND highest Airbnb demand
- S. Siro +96.5% — a peripheral neighbourhood with half the rent of Centro
- Baggio +65.9% — cheapest zone (€15.94/m²), modest Airbnb revenue still beats long-term rent

Chart 2: Choropleth Map

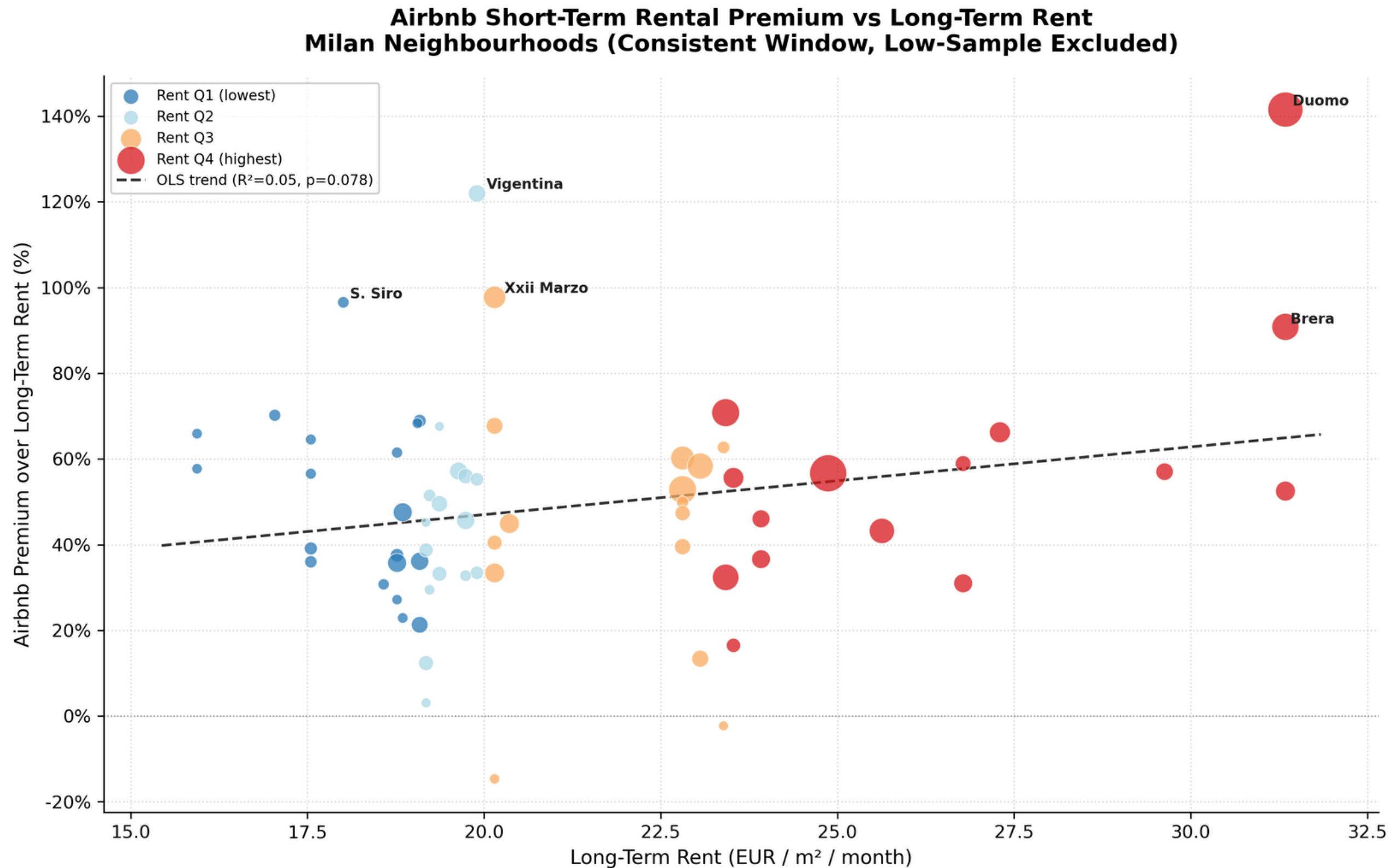
Short-Term Rental Premium by Milan Neighbourhood



→ **Live Demo Here!**

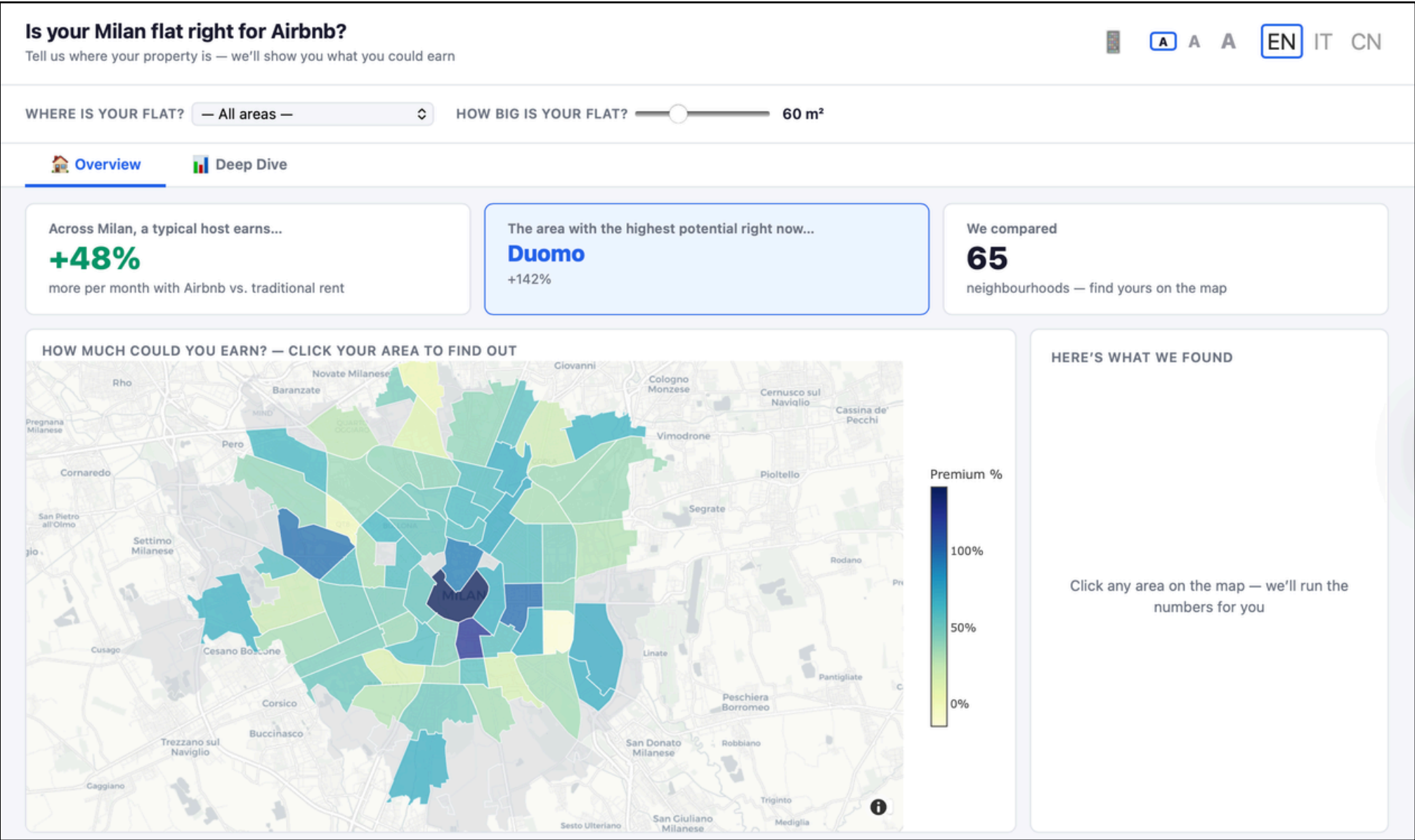
- Dark shading isn't only in the centre (Duomo/Brera)
- high-premium pockets also appear in the west (S. Siro), south (Vigentina, Rogoredo), and east (Xxii Marzo).
- Only one neighbourhood has no data: Cascina Triulza - Expo, which has zero Airbnb listings.

Chart 3: Scatter Plot



- Rent level explains only 5% of premium variation.
- Location type, not rent level, is what matters

INTERACTIVE DASHBOARD



We didn't trust Claude blindly...

Technical Checks

- manually recalculated premiums against raw CSV;
- Found a bug & fixed by adjusting the window;
- Switched mean to median

User Testing

- "Marco persona" prompt
- Tested with real Milan landlords
→ interface rated intuitive

"Maturity beats polish" — we caught the errors because we looked for them.

Three honest data limits

LIMITATION	WHAT IT MEANS	DIRECTION OF BIAS
 Asking rents, not transaction rents Source: Immobiliare.it	Listed rents may be higher than what tenants actually pay. The long-term rent baseline could be slightly overstated.	Premium understated
 Availability ≠ Occupancy Source: Inside Airbnb	<code>availability_365</code> records days open for booking — not days actually booked. Some listings may be open but underperforming.	Revenue may be overstated
 Zone-level LTR, not listing-level 31 zones → 87 neighbourhoods	Each of the 31 rent zones covers multiple neighbourhoods. Streets within the same zone may have very different actual rents.	Within-zone variation ignored

BUT...

These limitations define the right way to interpret the findings, they do not invalidate them

RECOMMENDATION

DO

- Use the dashboard before deciding
- Don't assume "city center = best"
- Adjust for your apartment size
- Check multiple neighbourhoods

INTERPRET CAUTIOUSLY

- Gross $\neq$ Net
- Legal restrictions vary
- Seasonality matters



What Does This Mean for Milan?

For Policymakers

- STR pressure is not just Duomo
- Consider neighbourhood-level regulation
- Monitor peripheral areas

For Future Research

- Net profit analysis needed
- Seasonality patterns
- Tenant displacement impact

Key Insights

Location beats property type – Where you are matters more than what you have



THANK
YOU

APPENDIX

THE 87 → 31 MAPPING TABLE

Centro (€31.34)

Brera, Duomo, Magenta - S. Vittore

Garibaldi Moscova Porta Nuova (€29.63)

Garibaldi Repubblica

Quadronno Palestro Guastalla (€27.30)

Guastalla

Arco Della Pace Arena Pagano (€26.78)

De Angeli - Monte Rosa, Pagano

Genova Ticinese (€25.63)

Ticinese

Porta Venezia Indipendenza (€24.87)

Buenos Aires - Venezia, Giardini Porta Venezia

Solari Washington (€23.92)

Tortona, Washington

Porta Romana Cadore Montenero (€23.53)

Porta Romana, Scalo Romana

Centrale Repubblica (€23.42)

Centrale, Loreto

Fiera Sempione City Life Portello (€23.39)

Parco Sempione, Portello, Qt 8, Tre Torri

Zona Navigli (€23.06)

Navigli, Parco Dei Navigli, S. Cristoforo

Cenisio Sarpi Isola (€22.81)

Dergano, Farini, Ghisolfa, Isola, Sarpi

Citta Studi Susa (€20.36)

Citta' Studi

Porta Vittoria Lodi (€20.15)

Corsica, Lodi - Corvetto, Ortomercato, Umbria - Molise, Xxii Marzo

Ripamonti Vigentino (€19.90)

Ex Om - Morivione, Ripamonti, Vigentina

Napoli Soderini (€19.74)

Giambellino, Lorenteggio, Selinunte

Maggiolina Istria (€19.64)

Maciachini - Maggiolina

Abbiategrosso Chiesa Rossa (€19.37)

Cantalupa, Gratosoglio - Ticinello, Parco Agricolo Sud, Parco Delle Abbazie, Quintosole, Stadera, Tibaldi

Famagosta Barona (€19.23)

Barona, Ronchetto Delle Rane, Ronchetto Sul Naviglio

Affori Bovisa (€19.18)

Affori, Bovisa, Bovisasca, Bruzzano, Comasina, Parco Nord

Precotto Turro (€19.09)

Greco, Padova, Viale Monza

Corvetto Rogoredo (€19.06)

Chiaravalle, Rogoredo

Bande Nere Inganni (€18.85)

Bande Nere, Forze Armate

Uptown Cascina Merlata Viale Certosa (€18.77)

Gallaratese, Maggiore - Musocco, Quarto Oggiaro, Sacco, Stephenson, Villapizzone

Udine Lambrate (€18.58)

Lambrate

San Siro Trenno (€18.01)

Parco Bosco In Citt, S. Siro, Trenno

Cimiano Crescenzago Adriano (€17.55)

Adriano, Parco Lambro - Cimiano

Bicocca Niguarda (€17.55)

Bicocca, Niguarda - Ca' Granda

Forlanini (€17.04)

Mecenate, Parco Forlanini - Ortica

Ponte Lambro Santa Giulia (€17.01)

Parco Monluè - Ponte Lambro, Triulzo Superiore

Bisceglie Baggio Olmi (€15.94)

Baggio, Figino, Muggiano, Quarto Cagnino, Quinto Romano

DATA CLEANING PIPELINE

Step	Records Before	Records After	Action Taken
Raw data	22,948	-	Initial dataset
Price conversion	22,948	20,553	Dropped 2,395 (10.4%) with missing prices
Outlier handling	20,553	20,553	Flagged 208 at p99 (€1,200/night) but retained
Final dataset	-	20,553	Used for neighbourhood-level analysis

Why Median, Not Mean?

Problem: Mean revenue was distorted by extreme listings.

Example: Parco Dei Navigli

Metric	Value
Listings	7
Mean monthly revenue	€41,694
Median monthly revenue	€3,570

Decision: Use median revenue for neighbourhood-level premium calculation. **Median is more robust**, especially when sample size is small.

PREMIUM CALCULATION

$$\text{Premium} = \frac{\text{Median Airbnb Revenue} - \text{Long-term Rent}}{\text{Long-term Rent}}$$

Airbnb Monthly Revenue

$$= \text{Price} \times \frac{\text{Availability}_{365}}{12}$$

Long-Term Rent (60 m²)

$$= \text{Rent}_{\text{EUR/m}^2/\text{month}} \times 60$$